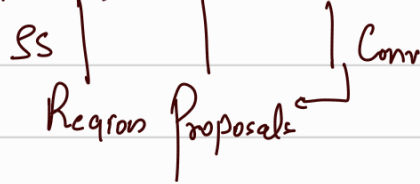
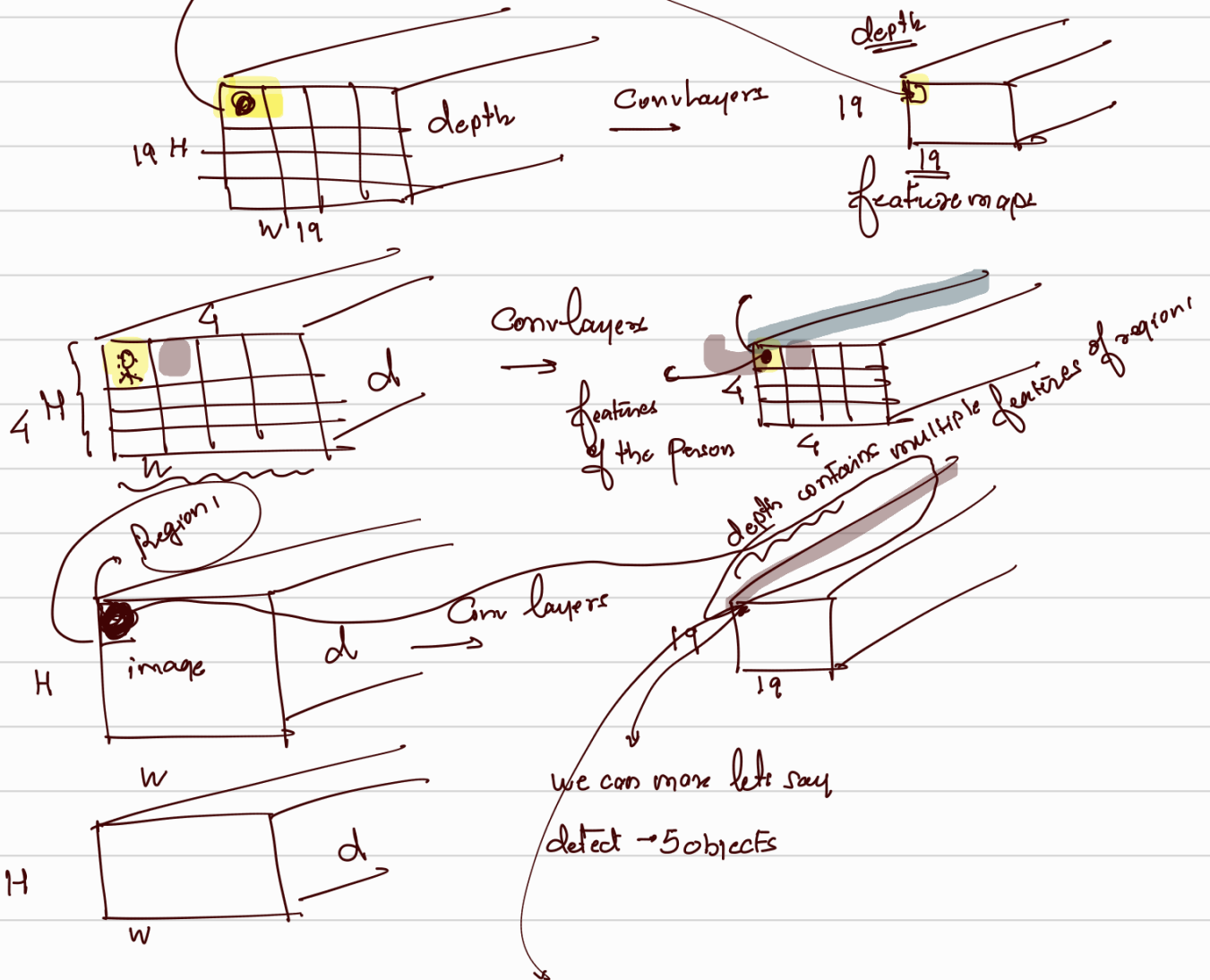


YOLO (You only look once) (Yohanes)

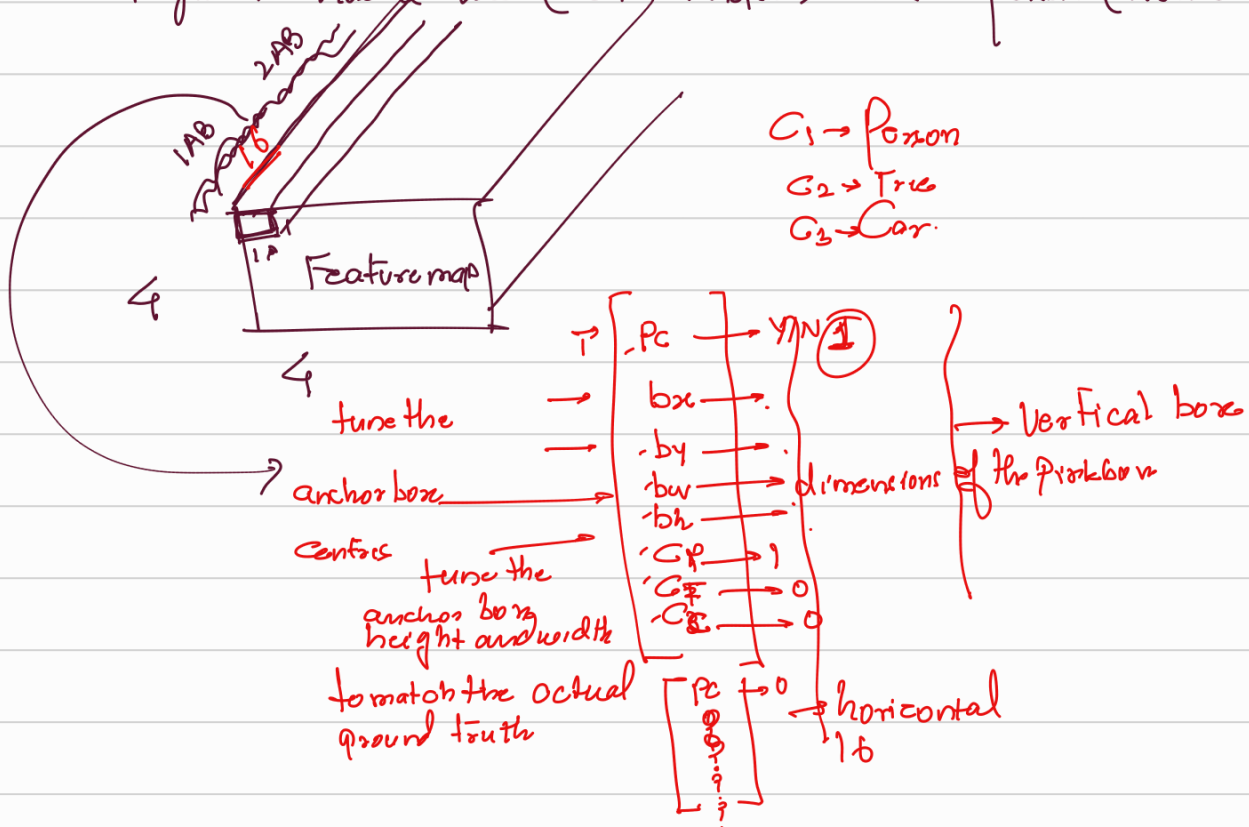
Object detection algorithms: R-CNN, Fast R-CNN, Faster R-CNN

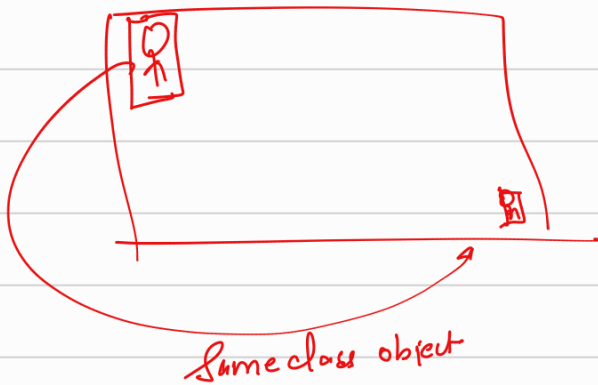


within the same network we detect both the region proposals and the bounding box



- $\rightarrow$ Anchor Box 1 $\rightarrow$ Anchor Box 5
- $\rightarrow$ Anchor Box 2 $\rightarrow$ Anchor Box 4
- $\rightarrow$ Anchor Box 3





Yolo-v1 could not match the capability to predict :

- ② overlapping objects
- ① same objects different size

Yolo-v2

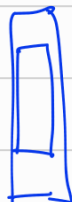
- ② Batch Normalization
- ② High Resolution images → Yolo-v1 → 224×224
4 448×448 and
- ② Anchor Boxes → we define them by doing K-means clustering on existing ground truths of my training data

② Feature map → 13×13

② 19 convolution layers and 5 more pooling layers

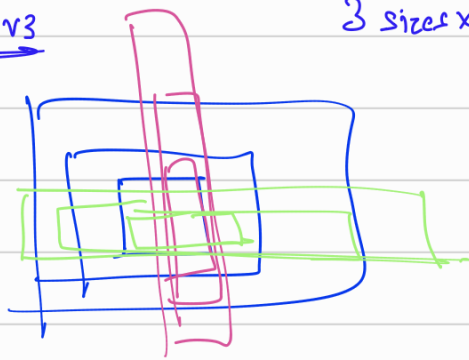
V2 faster, accurate

Yolo-v3 = 3×3



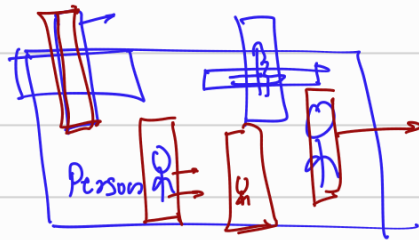
Yolo-v3

3 sizes x 3 aspect ratios



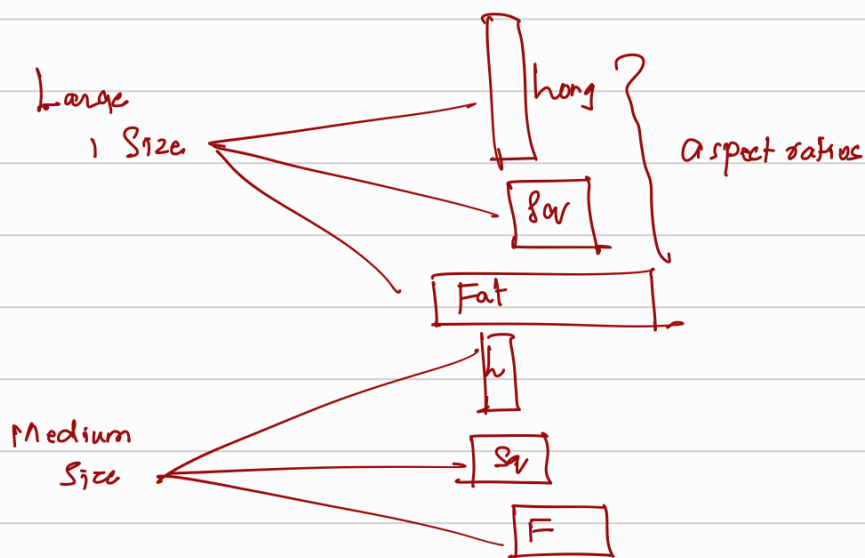
Anchor boxes $\rightarrow$ K-means Clustering

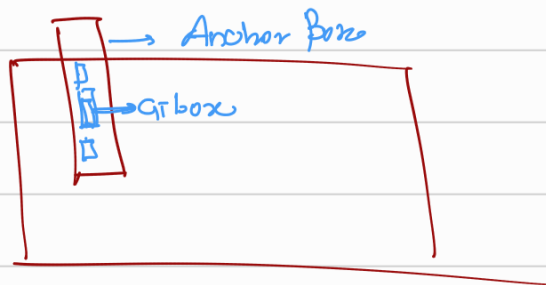
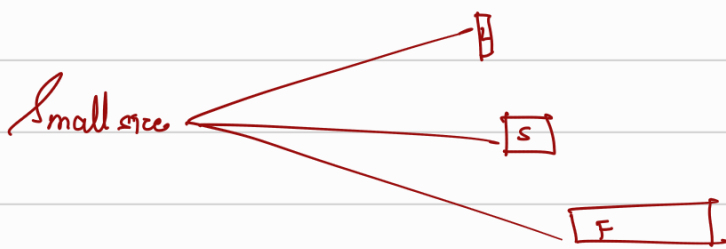
Yolo v3 model



Only one size for anchor boxes $\rightarrow$ 

Yolo-v3 $\rightarrow$ Size and aspect ratios for the anchor box

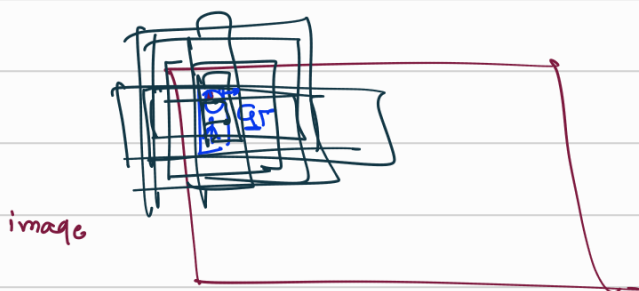
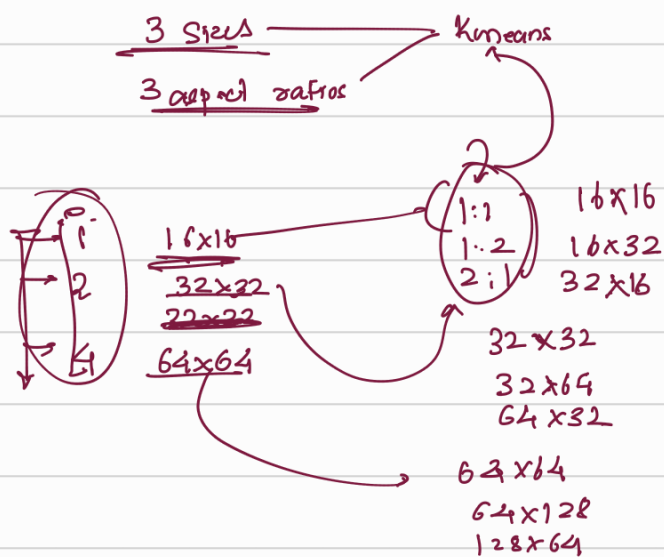




Yolov3 model $\rightarrow$ 3×3 Anchor Boxes

3 sizes, 3 aspect ratios

every grid cell will have 9 anchor boxes to fit in the ground truth.



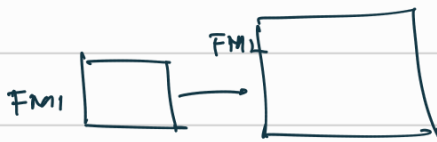
Assign the GT box to the most closest anchor box.



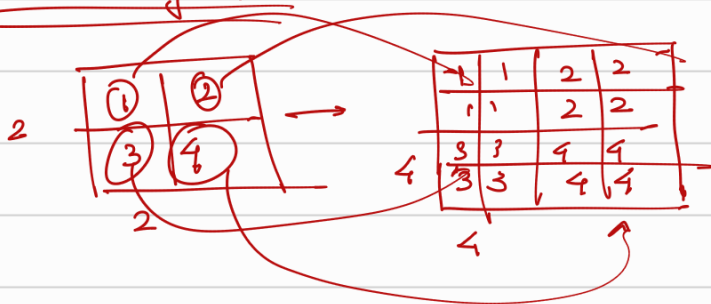
image $\rightarrow$ Convolutions $\rightarrow$ Downsample

FM $\rightarrow$ Convolutions $\rightarrow$ Bigger FM

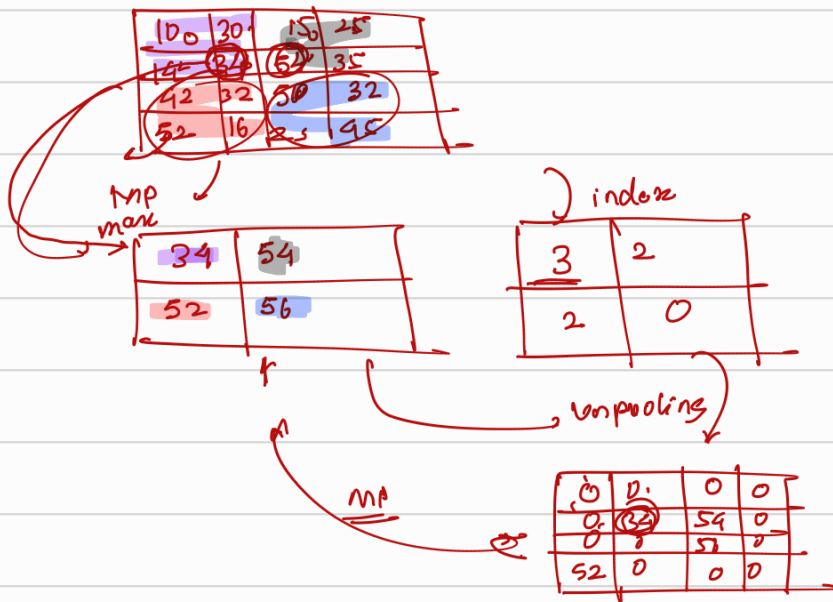
Upsampling (Transpose Convolution)



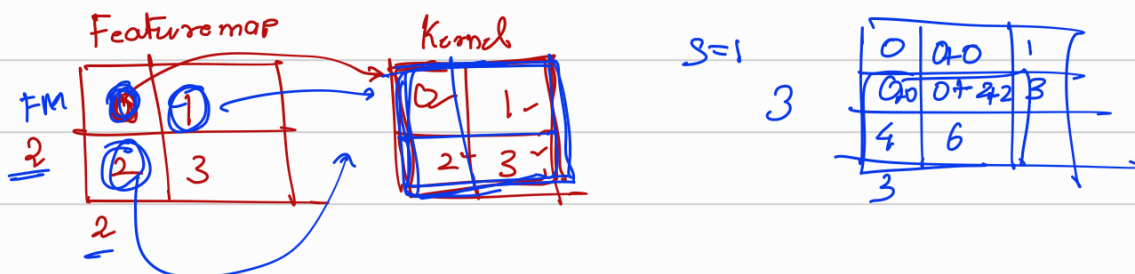
① Nearest neighbour

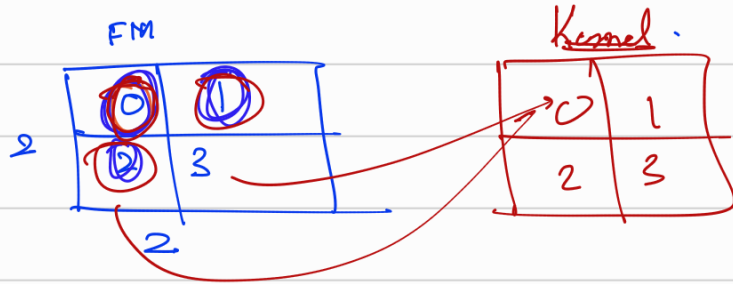
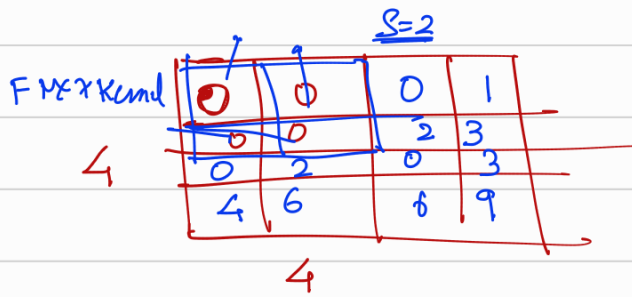


② Unpooling

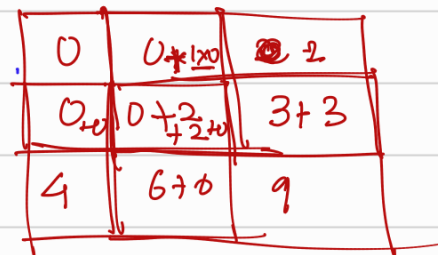


③ Transpose convolution

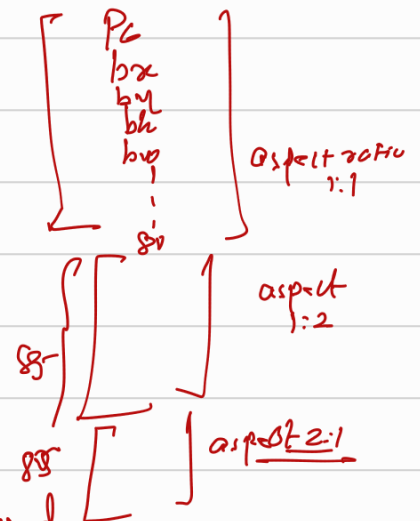
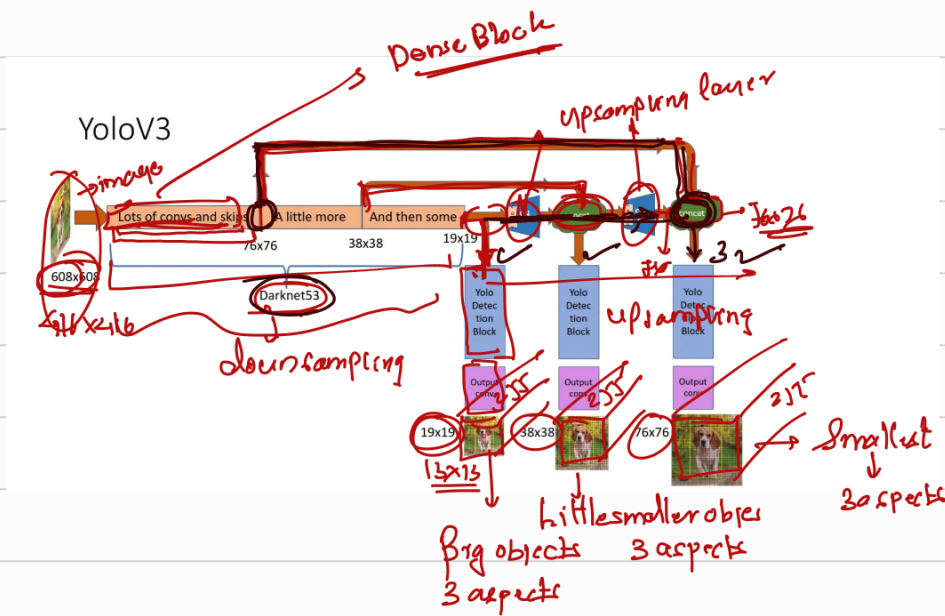




Output map



Transpose convolution is basically for up sampling your feature map



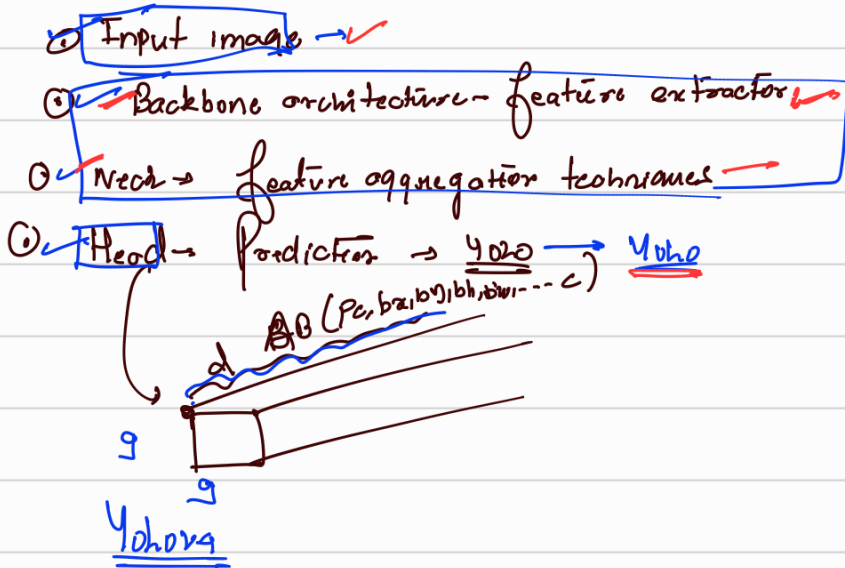
YoloV3 architecture is nothing but a series of convolutions applied along with certain skip connections to generate output at

19x19 (big object) → upsample → conv → 38x38 → upsample → 76x76

Yolo-v4 .. v7

Yolo v4 model

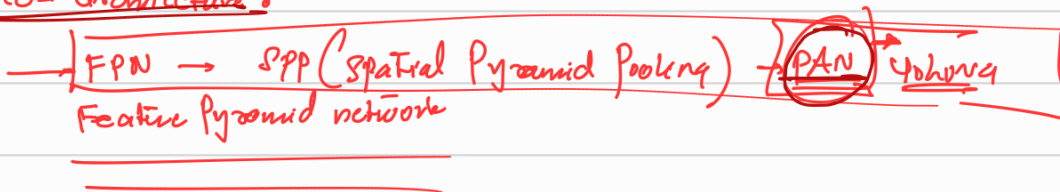
Building Components of an object detection technique:



Backbone architectures



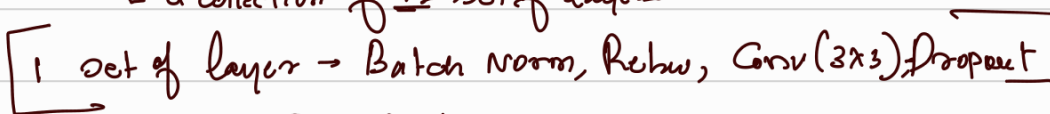
Neck architecture:



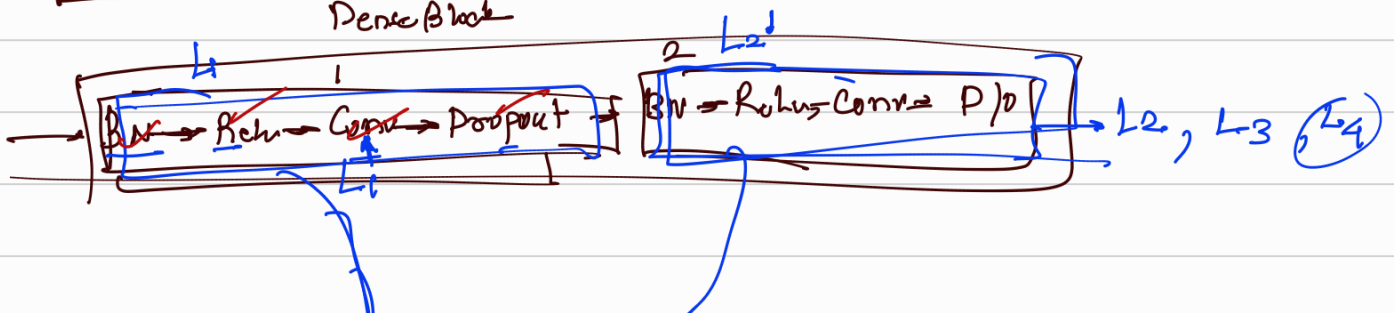
Dense Net (Backbone architecture)

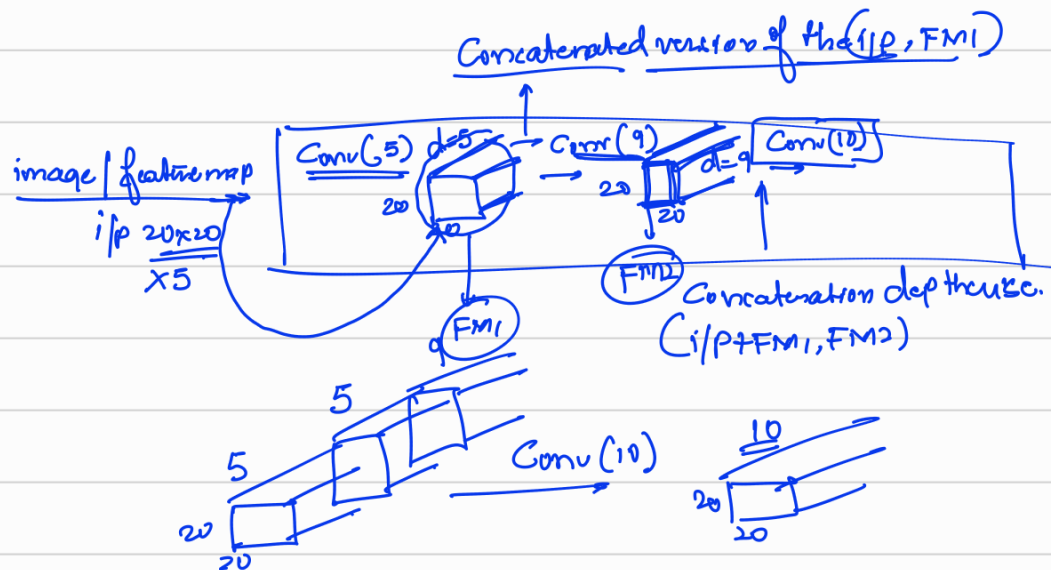
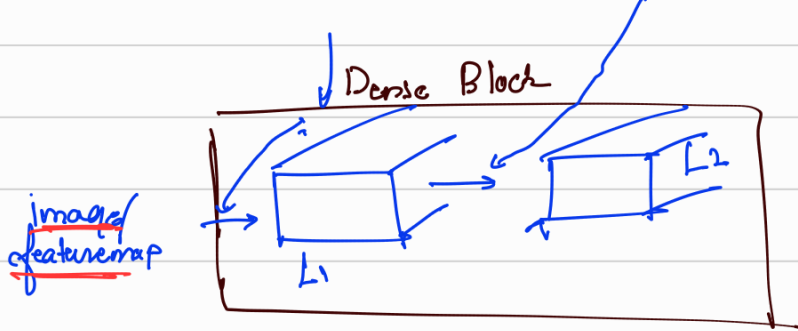


is a collection of a set of layers



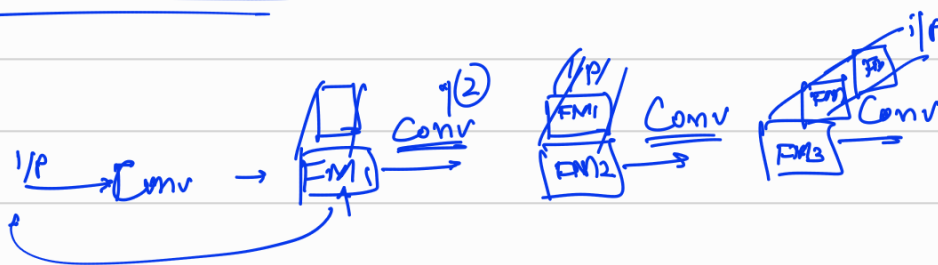
Dense Block





The dense block every time apply a Conv
we apply always to the concatenated version

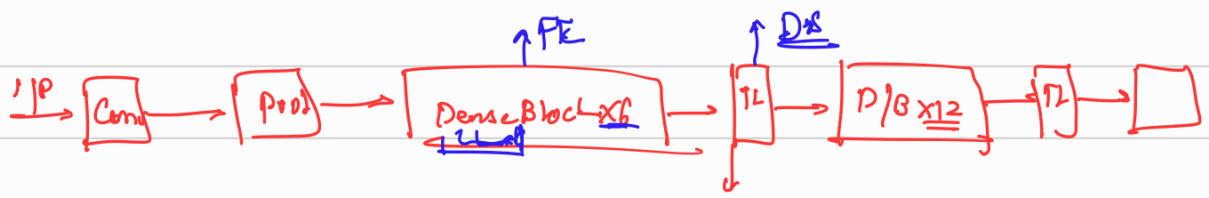
→ i^{th}
 $[FM_{i-1}, FM_{i-2}, \dots, i/p]$



this is a very computationally expensive phenomenon

Cross stage partial network

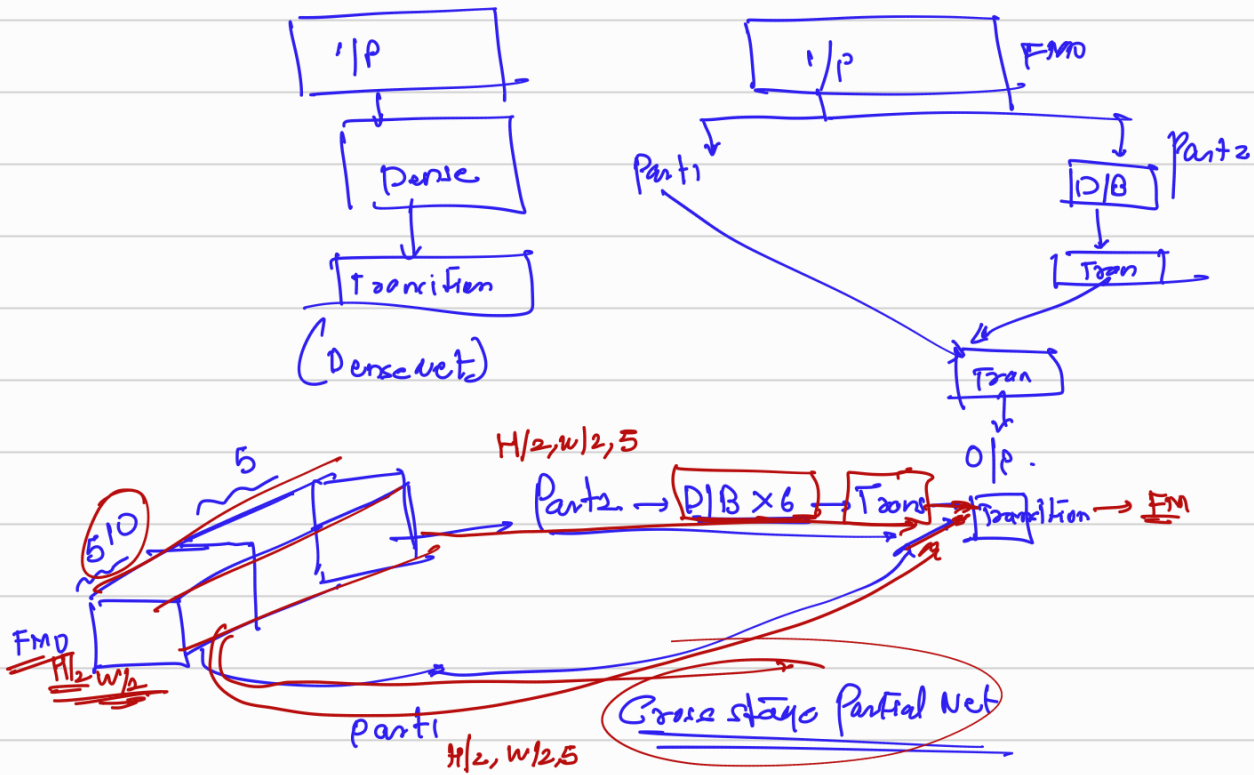
Transition Block



downsampling

TL $\rightarrow$ Batchnorm, ReLU, Conv(1x1), D/O, Pool(2x2)
 L₁

Gross Stage Partial Network



Yohov3

Mesh activation $\rightarrow$ Yohov4

Backbone Yohov3 $\rightarrow$ image $\rightarrow$... $\rightarrow$ d13 $\rightarrow$ d15 $\rightarrow$ Yoho

Backbone Yohov4 $\rightarrow$ image $\rightarrow$ d18 CSP $\rightarrow$ CSP $\rightarrow$ CSP
 [mesh activation]